

Differential Calculus — Chain Rule

Chapter 2: Power Rule on Cubic Polynomial | Page 33

PROBLEM STATEMENT

Power Rule & Cubic Composite

Find the derivative with respect to x of the function:

$$y = (x^3 + 2x - 1)^5$$

STEP 1 প্রদত্ত ত্রিঘাত বহুপদী ঘাত ফাংশনটি উল্লেখ করি (Given Composite Function)

$$y = (x^3 + 2x - 1)^5$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(x^3 + 2x - 1)^5]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^n] = nu^{n-1} \frac{du}{dx}$)

$$\frac{dy}{dx} = 5(x^3 + 2x - 1)^{5-1} \cdot \frac{d}{dx}(x^3 + 2x - 1) = 5(x^3 + 2x - 1)^4 \cdot \frac{d}{dx}(x^3 + 2x - 1)$$

STEP 4 বন্ধনীভুক্ত ত্রিঘাত অংশের অন্তরক নির্ণয় করি (Differentiating Inner Cubic Expression)

$$\frac{d}{dx}(x^3 + 2x - 1) = 3x^2 + 2(1) - 0 = 3x^2 + 2$$

STEP 5 মান বসিয়ে উৎপাদকীয় আকারে চূড়ান্ত অন্তরজ পাই (Final Factorized Form)

$$\frac{dy}{dx} = 5(3x^2 + 2)(x^3 + 2x - 1)^4$$

FINAL ANSWER
$$\frac{dy}{dx} = 5(3x^2 + 2)(x^3 + 2x - 1)^4$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 10

🔥 BOX 1: FORMULATION & POWER RULE ON POLYNOMIALS

- **Generalized Formula:** $\frac{d}{dx}[P(x)]^n = n[P(x)]^{n-1} \cdot P'(x)$.
- For $P(x) = x^3 + 2x - 1$, the inner derivative is $P'(x) = 3x^2 + 2$.
- With $n = 5$:

$$\frac{dy}{dx} = 5(3x^2 + 2)(x^3 + 2x - 1)^4$$

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Missing Parentheses:** Writing $5 \cdot 3x^2 + 2(x^3 + 2x - 1)^4$ instead of $5(3x^2 + 2)(x^3 + 2x - 1)^4$.
- **Attempting Full Polynomial Expansion:** Expanding a 5th power of a cubic polynomial results in 16 terms—always retain factored form in exams.

🌐 BOX 3: REAL DOMAIN & CRITICAL POINTS

- **Domain of Definition:** The polynomial function is defined and smooth on $\mathbb{R} = (-\infty, \infty)$.
- **Stationary Points Analysis:** $\frac{dy}{dx} = 0 \iff (3x^2 + 2)(x^3 + 2x - 1)^4 = 0$.
- Since $3x^2 + 2 \geq 2 > 0$ for all real x , real critical points can only come from the real root of $x^3 + 2x - 1 = 0$ (which has exactly 1 real root near $x \approx 0.4534$).

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = x^3 + 2x - 1 \implies y = u^5$.
- $\frac{dy}{du} = 5u^4$.
- $\frac{du}{dx} = 3x^2 + 2$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 5u^4 \cdot (3x^2 + 2) = 5(3x^2 + 2)(x^3 + 2x - 1)^4$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct power rule setup $5(x^3 + 2x - 1)^4$.
- **1.0 Mark:** Accurate computation of inner polynomial derivative $3x^2 + 2$.
- **1.0 Mark:** Final structured representation $5(3x^2 + 2)(x^3 + 2x - 1)^4$.